Diagnosing Spatial Policies in a Changing Commons

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Abstract

How does the value of spatial regulation change as ecological variability increases due to climate change? We develop a framework that decomposes the regulator's recursive problem into three structurally distinct efficiency conditions—intratemporal, spatial, and intertemporal—while allowing for spatial substitution, heterogeneous ecological shocks, and distributional concerns. Applied to detailed ecological and harvest data in the U.S. Atlantic sea scallop fishery, we find that the existing set of closures, harvest limits in certain areas, and day restrictions is too lax overall and has substantial spatial misallocation of harvests. Leakage destroys about 17% of the marginal value of future conservation. The regulations, meanwhile, implicitly undervalue the resource stock year to year. Climate-induced changes in the resource amplifies each type of inefficiency, though slowing mean growth versus increasing variance and spatial correlation of ecological shocks play varying roles. The existing regulation can be rationalized statically by rather moderate inequality aversion and dynamically by a regulator using out-dated expectations over the level and correlation of resource growth over space.

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1 Introduction

Regulators tasked with managing an ecologically important resource face significant spatial heterogeneity and substantial uncertainty over future biological growth, ecological shocks, and prices. Examples recur across environmental economics: a forest agency deciding which compartments to harvest and at what intensity under disturbances like wildfire and insect outbreaks; fire managers setting fuel-management strategies across large areas; a fishery manager setting quotas across spatial grounds where warming oceans change growth trajectories; a water board allocating extraction rights across basins under drought variability; an air-quality regulator pricing emissions across locations with spatially differentiated damages. In each case the underlying problem has the same structure: biology or geophysics is uncertain, the resource is unevenly distributed across cells, firms substitute their activity across space, and the environment is changing in ways that move multiple moments of the natural process simultaneously. Across all these commons problems, climate change makes future resource growth more uncertain, and the relationship between regulatory design and welfare becomes harder to characterize. This paper develops a framework for evaluating spatial environmental policy under uncertainty and an empirical exercise that evaluates a changing ocean's impact on the value of a spatial policy: the U.S. Atlantic sea scallop fishery.

How does the value of spatial regulation change as ecological variability increases? We set up tests based on an extension of the canonical natural resource problem with multiple areas, producers substituting across space, and ecological and economic uncertainty. It decomposes into three structurally distinct efficiency conditions, each diagnosable separately, each sensitive to a different feature of the environment, and each calling for a different reform. The first is an intratemporal condition and asks if there is inefficient harvest in that area: today's rent (the marginal profit at the restricted harvest i.e. the marginal cost of conservation) should equal the discounted future stock-value benefit of conserving the marginal unit (the marginal benefit of conservation). The second is a spatial condition across areas: at the optimum, the ratio of the marginal cost of conservation to the marginal benefit should be the same across space. The third is an intertemporal condition: the shadow value should evolve according to a modified Hotelling rule that allows for changes in harvest in one area to impact others (leakage). Importantly, as regulators set rules before uncertainty around resource growth and prices is realized, the conditions should be set ex-ante, considering a distribution of states, and allow for flexible social welfare functions by the planner. Climate projections that shift multiple moments together therefore produce qualitatively distinct failures of each condition, addressable only through correspondingly different reforms. One can evaluate the efficiency of a spatial policy under stationarity versus the observed changes in the resource over time.

We apply this framework to the U.S. Atlantic sea scallop fishery, among hte most valuable fisheries in the country and one of the most carefully monitored renewable-resource systems in the world. The fishery's institutional structure maps almost exactly onto the framework's primitives: patch-level harvest caps in rotating access areas, an aggregate days-at-sea (DAS) constraint that is not tradable across vessels, and full closures of patches whose biomass managers protect. The Northeast Atlantic is among the fastest-warming ocean regions in the world, generating shifting mean growth, growing marginal variance, and rising spatial covariance in the underlying biology. Trip-level vessel positions and landings, size-class-specific landings and prices, independent biomass surveys, and the assessment model's structural biology by management subarea together permit identification of each of the framework's diagnostics separately. We first document a set of reduced-form regularities that motivate the structural choices: vessels reallocate across space in response to regulatory status, with substitution measurable to 100–200 km and no detectable aggregate effect on harvest (consistent with the binding aggregate effort constraint); access-area caps on quantities bind tightly in practice while days-at-sea in open areas binds loosely; and harvest is strongly responsive to local biomass with no differential responsiveness across regulatory regimes. These regularities discipline the structural choices and pin down behavioral parameters that the counterfactual exercise relies on. We then specify a structural model of fleet choice and biology—a nested-logit choice model over (area, trip-day) combinations, size-structured stock dynamics with patch-specific recruitment shocks, and an estimated catch production function with decreasing returns to biomass—calibrated against the published NOAA assessment biomass and observed FY2014 area shares. Within this calibrated structure, a marginal-harvest perturbation experiment recovers the cap rent, the dynamic shadow value of stock, and the cross-patch leakage matrix at each access area. The actual closures, openings, and day/harvest limits are held fixed across the period.

The simulated perturbations allow us to test the three efficiency conditions. The static test suggests that, if the goal is to maximize harvest profits over time, the regulation is not strict enough: the marginal benefit of conservation (dominated by the precautionary value given very volatile biomass changes) exceeds the marginal cost by roughly 60% overall. This belies considerable spatial variation in the ratio across areas. The largest open-water grounds are sometimes under-harvested while several access areas are considerably over-harvested under existing rotations and rules, and the closed areas in 2014 are valuable enough that their closure is efficient for later profit increases. Opening closed areas earlier would incur considerable losses, and the existing regime clearly improves profits overall—the targeting of which areas to close versus limit-harvest is roughly correct, even though the level of fishing permitted is too lax. Leakage—the substitution of harvest to other areas when restrictions tighten—reduces the overall marginal benefit of conservation by about 17%, because the fleet operates under a binding aggregate days-at-sea constraint, so extra harvest in one patch crowds out effort elsewhere and preserves some stock there. The spatial misallocation is due either to the coarseness of the regulatory harvest restrictions or to the fact that harvest permits in access areas are not tradeable across boats. It turns out most of the spatial misallocation is from the former: approximately 73% of the spatial misallocation is between patches and only 10% between boats. Differentiating the allowed caps would close roughly seven times more of the spatial inefficiency than would making the existing caps tradeable across vessels—a priority ordering that is unusual relative to the standard fisheries-economics recommendation, which has historically emphasized tradable-quota reforms. Lastly, the dynamic test: by simulating shocks in the following year and re-estimating the marginal values of scallops in each area of the sea, one can see whether the regulation responds *ex ante* and roughly follows the Euler condition that would adjust stringency over time. This is rejected strongly under observed biomass growth, and the residual is strongly positive—the in-water stock is worth more than the regulator's revealed shadow value suggests. This could be due either to the already-too-lax static regulation or to a lack of updating to changing ecological conditions; to distinguish them we move next to the role of changing resource growth.

Welfare weighting changes the conclusions considerably. The static inefficiency falls rapidly at higher inequality preferences, and the current regulation is approximately statically optimal (the static wedge close to zero) for a planner with roughly inverse welfare weights (a log-linear social welfare function), reflecting moderate inequality aversion. This is because the current regulation opens up harvests in sensitive areas, and those profits—from large scallops that command high prices—accrue disproportionately to boats from the largest New England ports rather than to less profitable boats in the Mid-Atlantic. A planner who weights the latter more heavily sees the regulation's distributional pattern as approximately offsetting its static laxness. The dynamic residual also shrinks under inequality aversion—because the channel that drives it, mean-recruitment degradation, propagates through biology rather than through the distribution of fleet profits, so the weights it picks up are more muted—but the residual remains positive at every plausible distributional preference. This suggests that the regulator may willingly accept lower aggregate profits to shift harvest toward less profitable ports, but even at inequality preferences that rationalize the existing static restrictions in 2014, the regulation does not update enough to be robust to shocks going forward.

The central question is the role of changing resource growth under a changing ocean. We hold fixed the mean growth, variance, and covariance of biomass shocks at their early-period values and re-estimate the marginal benefits and costs against the late-period values, then decompose how each moment contributes. First, changing ecological conditions aggravates aggregate static inefficiency, mostly driven by growing variance and increased spatial correlation of biomass shocks rather than by slowing mean growth. The mechanism is precautionary: more variable and more synchronized shocks raise the buffer value of in-water stock, because leaving scallops in the water creates an option to harvest them when they grow into the highest-priced grades and a hedge against bad realizations. The changing resource also makes spatial inefficiency considerably worse, driven mostly by changes in mean growth in the Mid-Atlantic, where scallop beds have stopped growing nearly as large and existing regulations remain especially lax (the increasing spatial correlation actually works to slightly *reduce* spatial inefficiency, because more synchronized shocks make patches' precautionary responses more uniform). Lastly, and most strikingly, dynamic inefficiency is almost entirely driven by changing resource growth: under stable growth parameters and a stable spatial correlation matrix, the dynamic-efficiency test is indistinguishable from zero. We cannot reject that the spatial policy changes over time roughly in line with maximizing value—albeit a value constrained by the inefficient spatial allocation we documented above. This is despite considerable allocative inefficiency inherent in the non-tradeable harvest restrictions and non-tradeable day allocations. Most of the dynamic change is driven by slowing mean growth rather than variance or correlation: the rotating closures are not responsive enough to the considerable slowing in overall scallop biomass.

Future work will consider how far the existing instruments—closures, changing access limits, day restrictions—can be adjusted to increase efficiency across the three tests. Then we will turn to more ambitious regulatory reforms, ones that might manage leakage better and others that are robust to the non-stationary nature of the resource: what designs maintain resource value *ex ante* across the range of climate trajectories? This question pervades environmental regulation and can inform regulatory design and instrument choice across resource issues.

Our three-FOC diagnostic structure can be applied to any setting in which a recursive regulator faces spatial heterogeneity under uncertainty—forest management with stochastic growth, pollution permits under uncertain damages, water rights under hydrological variability. The variance-decomposition methodology for reform priority applies whenever multiple instruments can address related but distinct misallocations and a quantitative ranking is required. The static-dynamic decoupling under welfare weighting is a general empirical regularity in stochastic dynamic settings: when the static wedge is dominated by precautionary covariance terms aggregating over agent-level profits and the dynamic residual is driven by mean-process channels propagating through the costate, the two move differently under welfare weighting.

Related literature. Our contribution is threefold. First, we develop a framework that decomposes the regulator's recursive optimization into three structurally distinct efficiency conditions—intratemporal, spatial, and intertemporal—each diagnosable separately and each mapping to a different reform when violated. Second, we provide empirical evidence on spatial substitution, binding constraints, and the cross-vessel and cross-patch structure of misallocation in a major rotational management program. Third, we use a structural model to diagnose each efficiency condition, decompose its sensitivity to climate and to distributional preferences, and rank competing reform levers through a variance decomposition of the underlying misallocation.

The paper builds on foundational work in renewable-resource economics on optimal harvesting under uncertainty (Reed 1979; Clark 1990; Pindyck 1984; Weitzman 2002; Pizer 2002), which established the precautionary value of in-water stock, and on the robust-control tradition (Hansen and Sargent 2008; Brock and Hansen 2018). We build on these by separating the dynamic problem into its three constituent first-order conditions, each empirically distinguishable: when the environment is non-stationary, single-Euler treatments lose identifying power and structurally finer decompositions deliver sharper policy implications.

The paper also extends classic work comparing price and quantity instruments under uncertainty (Weitzman 1974; Newell and Pizer 2003; Costello and Karp 2004). Our contribution is to show how these tradeoffs are transformed in spatial settings where producers reallocate activity across locations and multiple instruments coexist. The between-cell vs. between-agent variance decomposition we develop provides a quantitative ranking of reform levers—something the canonical price-vs-quantity formulation does not confront because it lacks spatial heterogeneity.

We contribute to the empirical literature on fisheries management, which evaluates catch shares (Costello et al. 2008), effort limits (Essington et al. 2012), and spatial management tools (White and Costello 2014), often focusing on average effects. We instead produce a diagnostic structure for which margins the regulation actually fails, and complement work on the distributional impacts of environmental policy (Ryan 2019; Sudarshan 2020) by showing that the directional reading of the regulation's static-efficiency failure is planner-specific while the dynamic-efficiency failure is not.

We connect to literature on spatial policies under displacement and spillovers (Reynaert, Souza-Rodrigues, and van Benthem 2023; Earle Grupp et al. 2026; Souza-Rodrigues 2018; Assunção, Gandour, and Souza-Rodrigues 2019; Assunção et al. 2022). Relative to this work, our focus is not on whether a particular set of locations is selected for protection but on how a given regulatory regime maps shocks into realized economic outcomes through the three efficiency conditions.

Our framing is motivated by climate risk in natural-resource systems. Climate change alters expected ecological productivity, variance, and spatial correlation of ecological shocks in spatially differentiated ways (Pinsky and Mantua 2014; Free et al. 2019; Mills et al. 2013). Our framework makes the link between these moments and the three efficiency conditions explicit. Closely related is the portfolio view of environmental policy (Mallory and Ando 2014); rather than evaluating portfolios holistically, we identify which condition a regulation fails, attribute the failure to a specific moment of the shock distribution, and map to a specific reform lever.

2 Framework: spatial regulation under uncertainty

We model a regulator who manages a stochastic, spatially heterogeneous renewable resource using a coarse policy mix—location-specific harvest caps together with an aggregate effort allowance—facing a mass of extractors who respond optimally to the policy. The resource evolves under shocks whose distribution shifts with the environmental state. The framework has the standard structure of a dynamic stochastic resource problem; we develop the parts that produce the three diagnostic conditions the empirical exercise tests. Full derivations, the size-structured generalization, and the welfare-weighted extension are in Appendix A.

2.1 Setup

Discrete time $t = 0, 1, \ldots$. The resource is distributed over K spatial cells ("patches") indexed by $k = 1, \ldots, K$. The stock vector is $X_t = (X_{1,t}, \ldots, X_{K,t})$ and evolves as $X_{k,t+1} = G_k(X_{k,t}, q_{k,t}, \varepsilon_{k,t+1})$, with $G_{k,X} > 0$ (biological growth), $G_{k,Q} < 0$ (harvest depletion), and $\varepsilon_{t+1} \sim F_\varepsilon(\cdot; \theta_t)$ a vector of patch-level shocks. The distribution F_ε depends on an environmental state θ_t whose mean, marginal variance, and cross-patch covariance characterize the underlying environment; the trajectory of θ_t over time encodes how the environment changes.

The regulator chooses a policy $I_t = (\{\bar{Q}_{k,t}\}_{k\in\mathcal{A}}, \bar{T}_t)$, where $\bar{Q}_{k,t}$ is a per-patch harvest cap at locations in a designated set $\mathcal{A}$ (call them *access* areas) and $\bar{T}_t$ is an aggregate effort allowance distributed across the extractor population. Locations outside $\mathcal{A}$ are either open (harvest constrained only by aggregate effort) or fully closed (harvest prohibited). Extractors take the policy and the stock as given and choose harvest $q_t = (q_{1,t}, \ldots, q_{K,t})$ to maximize per-period profit $\sum_k \pi_{k,t}(q_k, X_{k,t}) = \sum_k \left[p_t q_k - C_k(q_k, X_{k,t})\right]$ subject to the per-patch caps and the aggregate-effort budget, with the harvest cost function satisfying $C_{k,q} > 0$, $C_{k,qq} > 0$, and $C_{k,X} < 0$ (lower harvesting cost when stock is abundant). Let $T_k(q_k)$ be the effort cost of harvesting q_k at patch k.

2.2 The regulator's recursive problem

Let $W(X_t, I_t)$ be the within-period welfare flow under policy I_t. In the baseline equal-weighted treatment, $W(X_t, I_t) = \sum_k \pi^*_{k,t}(X_t, I_t)$ is aggregate extractor profit. The regulator chooses the sequence of policies $\{I_t\}$ to maximize expected discounted welfare, with value function

$$V(X_t) = \max_{I_t} \left\{ W(X_t, I_t) + \beta\, \mathbb{E}_t\left[V(X_{t+1})\right] \right\}, \tag{1}$$

where $\beta \in (0,1)$ is the discount factor.

The extractor's response. For each cap configuration I_t, extractors choose $\{q^*_k\}$ satisfying the first-order condition

$$p_t = C_{k,q}(q^*_k, X_{k,t}) + \xi_{k,t} \cdot 1[k \in \mathcal{A}] + \mu_t\, T'_k(q^*_k), \tag{2}$$

with $\xi_{k,t} \geq 0$ the multiplier on the per-patch cap (the *cap rent*: the marginal value to extractors of cap relaxation by the envelope theorem) and $\mu_t \geq 0$ the multiplier on the aggregate-effort budget (the *effort rent*). These two shadow values are the central state-contingent objects the empirical exercise recovers from observed behavior.

2.3 The static condition at each patch

Differentiating the Bellman (1) with respect to a patch-level cap $\bar{Q}_{k_0,t}$ at access area $k_0 \in \mathcal{A}$ and applying the envelope theorem to the within-period welfare yields the first first-order condition:

$$\boxed{\xi_{k_0,t} = -\beta\, \mathbb{E}_t\left[\sum_j \lambda_{j,t+1} \cdot G_{j,Q} \cdot \frac{\partial q^*_j}{\partial \bar{Q}_{k_0,t}}\right],} \tag{3}$$

where $\lambda_{j,t+1} \equiv V_{X_j}(X_{t+1})$ is the *costate*: the shadow value of stock at patch j next period, the regulator's implicit marginal value of an additional unit of in-water stock. Equation (3) is the patch-level intratemporal balance condition—the marginal cost of conserving stock today (the cap rent the extractors would forgo) should equal the marginal benefit of conserving stock (the discounted future stock-value benefit). We refer to it as the "$\mathrm{MC} = \mathrm{MB}$" condition.

Leakage and the cross-patch coupling. The right-hand side of (3) sums over *all* patches j, not just the patch k_0 being perturbed. The reason is that relaxing the cap at k_0 changes harvest at every other patch through the implicit function theorem applied to (2). At k_0 itself, $\partial q^*_{k_0}/\partial \bar{Q}_{k_0,t} = 1$ trivially. For $j \neq k_0$, more harvest at k_0 consumes more aggregate effort, tightening the effort budget and raising μ_t; this in turn reduces harvest at other patches through (2). The full cross-patch derivative is derived in Appendix A.2; the substantive observation is that

$$\frac{\partial q^*_j}{\partial \bar{Q}_{k_0,t}} \propto -\frac{\partial \mu_t}{\partial \bar{Q}_{k_0,t}},$$

so the cross-patch derivatives are negative under binding aggregate effort and *identically zero* if the aggregate-effort constraint does not bind. Leakage is generated by aggregate-effort coupling: the magnitude of cross-patch substitution is due to common downward-sloping demand for the resource across space, or (even when prices are fixed) due to regulatory environments where effort allocations are non-tradeable or otherwise generate a binding fleet-wide constraint

Applying $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y] + \mathrm{Cov}(X,Y)$ to (3) decomposes the right-hand side cleanly:

$$\xi_{k_0,t} = \underbrace{\beta\, \mathbb{E}_t[\lambda_{k_0,t+1}]\, |G_{k_0,Q}|}_{\text{(a) own-patch saving}} \underbrace{- \beta \sum_{j \neq k_0} \mathbb{E}_t[\lambda_{j,t+1}]\, |G_{j,Q}| \left| \frac{\partial q^*_j}{\partial \bar{Q}_{k_0,t}} \right|}_{\text{(b) leakage}} + \underbrace{\beta \sum_j \mathrm{Cov}_t\!\left(\lambda_{j,t+1},\, -G_{j,Q} \cdot \frac{\partial q^*_j}{\partial \bar{Q}_{k_0,t}} \right)}_{\text{(c) precautionary covariance}}. \tag{4}$$

Term (a) is what the conservation cost would be in a single-patch, deterministic world. Term (b) is the across-patch reallocation: relaxing the cap at k_0 preserves stock at other patches in expectation, partly offsetting the conservation cost. Term (c) is the precautionary covariance and is identically zero under deterministic biology; it captures the second-moment buffer value of in-water stock and is where the variance and spatial covariance features of θ_t enter the wedge.

2.4 The costate equation

The second first-order condition comes from differentiating the Bellman (1) with respect to the state $X_{k,t}$. The regulator has optimized I_t, so indirect effects through the policy vanish by envelope, and the law of motion for the costate is

$$\boxed{\lambda_{k,t} = -C_{k,X}(q_k^*, X_{k,t}) + \beta\,\mathbb{E}_t\left[\sum_j \lambda_{j,t+1}\cdot\left(G_{j,X_k} + G_{j,Q}\cdot\frac{\partial q_j^*}{\partial X_{k,t}}\right)\right].} \tag{5}$$

The first term is the contemporaneous cost saving from an extra unit of stock at k; the second is the recursive component, propagating forward through biology persistence (G_{j,X_k}) and through extractors' response to current stock.

Equations (3) and (5) are structurally distinct. The first is a within-period optimality condition for each patch separately; the second is a law of motion for the shadow value of stock across periods. The standard Euler treatment collapses the two by substituting (5) into the future term of (3); the empirical exercise in this paper instead tests them separately.

2.5 Three diagnostic conditions

Three structurally distinct diagnostics of regulatory efficiency follow from (3) and (5).

Static intratemporal efficiency requires (3) to hold at each access area. Empirically, the wedge $\mathrm{MC}_k - \mathrm{MB}_k$ at each patch tests this, with MC_k approximating $\xi_{k,t}$ by the gross dock value of the marginal harvest and MB_k computing the right-hand side by simulating the forward consequences of a marginal harvest. A positive wedge ($\mathrm{MB} > \mathrm{MC}$) means the regulation permits more harvest than the dynamic optimum would.

Static spatial efficiency requires (3) to hold at *every* access area simultaneously. At the spatially-efficient optimum, the marginal benefit per unit of in-water stock should be equalized across active patches—if any patch has MB per kg materially higher than another, the regulator could reallocate harvest pressure between them and raise welfare. The diagnostic is dispersion in the ratio of MB to MC across patches.

Dynamic intertemporal efficiency requires (5) to hold along the regulated path. Equivalently, the realized residual

$$u_{k,t+1} \equiv \beta\big(-C_{k,X} + \lambda_{k,t+1}\big)(1 + G_{k,X_k}) - \lambda_{k,t}$$

satisfies $\mathbb{E}[u] = 0$ under the dynamic optimum, in the spirit of GMM tests of Euler conditions in stochastic dynamic settings (**?**). A positive $\mathbb{E}[u]$ means the regulation treats the resource as less scarce than its forward asset-pricing value implies, given the shocks the resource has been experiencing.

Each condition is necessary for full dynamic optimality, and each can fail independently of the others. Each is sensitive to a qualitatively different feature of θ_t: the static intratemporal wedge moves through the second-moment buffer value (term (c) in equation 4), responding to variance and spatial covariance of biology shocks; the spatial dispersion across patches responds to the joint structure of all three moments; the dynamic residual moves through the mean of θ_t via the persistence factor in (5). The empirical exercise in Section 6 tests each condition separately and attributes climate-induced movement in each to a specific channel.

2.6 Where inequality aversion enters

The baseline aggregator $W = \sum_i \pi_i^*$ weights all extractors equally. A welfare-weighted generalization replaces it with $W = \sum_i w_i \, U(\pi_i^*)$ under a concave utility U and weights w_i chosen by the planner. Under CRRA with parameter p, $U'(\pi) = \pi^{-p}$ and the natural weights are $w_i \propto \text{revenue}_i^{-p}$, placing more weight on lower-revenue agents as p rises.

Both first-order conditions pick up these U'-weights, but they do so asymmetrically across the two conditions, and this asymmetry has substantive content. The Control FOC (3) and the decomposition (4) reweight by the marginal utility of profits at patches where conservation gains concentrate—in a setting where those gains accrue disproportionately to high-revenue agents, an inequality-averse planner shrinks the wedge and the static-efficiency conclusion becomes planner-specific. The Costate equation (5) reweights symmetrically through $-C_{k,X}$ and the forward $\lambda_{j,t+1}$, both of which propagate through the biology process rather than through the cross-agent distribution of profit; the dynamic residual under welfare weighting shrinks in magnitude but tends not to flip sign. The static and dynamic conditions therefore *decouple* along the equity dimension—the static case is planner-specific while the dynamic case is planner-invariant in sign. This decoupling is a property of the two FOCs taken together, not of the empirical application; Section 6 documents it quantitatively at plausible inequality-aversion parameters.

2.7 Extensions

We highlight two additional details for empirical applications. The first is harvester heterogeneity: the framework above uses a representative agent, but the per-agent FOC (2) and the leakage matrix generalize to a heterogeneous population by indexing costs, effort allowances, and effort shadows by harvester. Aggregate $\xi_{k,t}$ and μ_t become weighted aggregates of agent-level shadows. This generalization allows the variance decomposition in Section 6 to separately identify the between-agent component of the inefficiency (e.g. driven by non-tradability of harvest allocations) from the between-patch component (driven by the coarseness or level of the per-patch restrictions). Marginal profits at the harvester level can still be aggregated to give marginal profits at the patch level.

Second, quality structure: many renewable resources mature into higher-value classes over time, and the within-period profit function depends on the size or quality composition of the catch rather than just on the harvest mass. In addition, production technologies are such that it is only possible to select areas to harvest rather than target particular classes (sawtimber versus pulpwood or large versus small fish). Replacing the scalar X_k with a vector over quality classes turns the costate equation into a matrix recursion to capture maturation between classes. The own-patch term in the decomposition (4) gains an explicit *maturation premium*: the shadow value of a marginal small unit is augmented by the discounted probability that it matures into the higher-priced class. It leads to a hump-shaped impulse response of profit to marginal harvest changes, with the loss peaking one to two years after the perturbation when the harvested cohort would have entered the highest-priced class. Derivations and operational expressions for both extensions are in Appendix A.

3 Setting and Data

The U.S. Atlantic sea scallop fishery is the highest-revenue fishery on the U.S. East Coast and operates along the Northeast continental shelf from the Mid-Atlantic Bight to Georges Bank. The fishery has three features that make it a stringent test setting for the framework. First, regulation maps almost exactly onto the framework's primitives: a limited-access fleet receives non-tradable annual "days at sea" (DAS) allocations and is subject to patch-level harvest caps in rotating access areas, with additional patches fully closed to protect biomass. Second, fleet substitution across spatial patches is observable in detail: trip-level vessel positions and landings are recorded for the full fleet from 2008 to 2022, with size-class-specific landings (the under-10 "u10" class commanding prices roughly an order of magnitude above the smallest grades) and detailed homeport and vessel characteristics. Third, the Northeast Atlantic is among the fastest-warming ocean regions in the world, with seafloor temperature shifts, recurring marine heatwaves (including a major event in 2012), and growing prevalence of gray-meat disease—generating shifts in mean recruitment, marginal variances, and cross-patch covariance in the underlying biology.

Two features of the regulatory instrument are important as sources of misallocation. The DAS allocation is non-tradable across vessels: boats whose DAS constraint binds tightly cannot acquire effort from boats whose constraint is slack, which is the source of the empirical leakage matrix (Section 2.3) and the between-boat dispersion that an ITQ-style reform would compress. The access-area cap structure is spatially coarse: caps are set patch-by-patch by year, without fine-grained differentiation by underlying biological shadow value. A patch-level cap structure that responded more finely to local shadow value would compress a different wedge. Section 6.1 quantifies both and delivers a sharp priority ordering. The Fisheries Management Council that sets these caps does so in advance of realized biology and prices, with the published Plan Development Team (PDT) Scallop Area Management Simulator (SAMS) providing biomass and size-distribution projections by management subarea; we use the SAMS panel directly in the structural exercise.

Climate change shifts *three* structurally relevant features of the biology process—mean recruitment, marginal variance of growth shocks, and the cross-patch covariance of those shocks—and the existing regulatory rule responds to none of them in a state-contingent way. The Mid-Atlantic, the southernmost and warmest part of the fishery, faces the most considerable change: recruitment success has become more sensitive to local conditions across the range, and projections indicate continued warming with rising heatwave risk. Climate change changes the risk profile under which spatial policy operates, shifting the distribution and comovement of biological outcomes across space and time, which in turn changes the welfare value of different spatial regulatory configurations along each of the three efficiency conditions.

Data. We use NOAA Fisheries trip records (2008–2022), which contain daily landings by patch and size class with detailed vessel characteristics, location and timing of fishing activity, and homeport identifiers. These records are matched to NOAA biomass surveys generating spatially explicit estimates of scallop abundance and size structure, and to the rotational management system's record of which patches were closed, access, or open in each year (built from public NOAA shapefiles and digitized polygon coordinates from Federal Register announcements). Figure 1 maps the distribution of harvests and the constructed patch boundaries used in the location-choice model.

The structural model requires a patch-level, size-structured panel of in-water biomass. We construct this from the published PDT/SAMS combined-survey biomass estimates by management subarea, mapping each model patch to its nearest SAMS subarea and allocating the published total across patches by where the fleet actually fishes. This construction is one of three foundational data fixes the empirical exercise relies on (the other two are the baseline-allocation calibration in Section 5 and the marginal-harvest instrument in Section 5). It delivers patch-level biomasses consistent with the published PDT/SAMS totals and fleet exploitation rates ($F \approx 12\%$) matching the assessment, in contrast to earlier survey-sliver constructions that produced order-of-magnitude inconsistencies between patch biomass and observed catch on open-water patches. Validation is in Appendix **??**. Detailed institutional history of the fishery, the regulatory timeline since 1982, the rotational closure schedule, the structure of the Council process, and the auction-based price formation by meat-count grade are described in Appendix **??**.

4 Reduced-form evidence

We document reduced-form regularities that motivate the structural choices in Section 5 and that each of the framework's structural objects will need to reproduce. Summary statistics on the sample are in Table 1.

Vessel choice responds strongly to regulatory status, on the extensive margin. A conditional logit of trip-level location choice (Table 2) on the regulation-varying choice set finds that closure reduces the probability that an area is selected by approximately 49% relative to open areas, while access designation nearly doubles selection probabilities, conditional on a precisely estimated distance penalty. Decomposing into extensive and intensive margins (vessel-area-year participation logit and Poisson trip count conditional on participation), access designation increases the probability that a vessel fishes in an area by about 82%, while conditional trip counts barely move (a 2% point estimate, indistinguishable from zero). Closures cut participation by $\sim 69\%$ but leave conditional intensity unchanged. The behavioral channel through which spatial regulation operates is whether vessels show up at a patch, not how heavily they fish once present—a fact that disciplines the choice-diffuseness parameter in the structural model.

Spatial substitution reaches 100–200 km, with no aggregate effect. A stacked difference-in-differences across regulatory events (closures, openings, access-area designations), comparing 0.1° cells in concentric distance bands from the affected patch, finds that closures generate sharp declines in activity within the treated patch with offsetting increases in nearby patches—largest in the 0–15 km band and statistically detectable to 100–200 km. Aggregate activity across bands is essentially unchanged: the fleet redistributes effort, it does not reduce or increase total effort, consistent with the binding aggregate DAS constraint that gives the framework's leakage matrix a non-trivial structure. Openings show mirror-image patterns. Figure 2 reports the event-study estimates.

Access-area caps bind tightly; DAS in open areas binds loosely. The distribution of annual trips to access areas, normalized by the modal regulatory threshold, exhibits a spike-to-baseline ratio exceeding $20\times$ at the limit, with similarly sharp bunching at the annual access-area harvest cap (Figure 3). The DAS distribution in open areas is smooth around its threshold with only modest excess mass at the cap. Two features of the institutional setting contribute: measurement error in recorded DAS (estimated from "days absent" rather than directly observed) and the empirical observation that vessels routinely do not exhaust their annual DAS allocation. The non-tradable structure means the aggregate constraint still binds at the vessel level for the high-shadow subset of the fleet ($\sim 60\%$); the cap shadow $\xi_k > 0$ assumed in the structural exercise is directly confirmed in the access-area trip distribution.

Harvest is responsive to local biomass, with no regime differentiation. Estimating year-over-year elasticity of harvest to biomass at the patch level gives a baseline elasticity of 1.55 (s.e. 0.18) and an interaction with the open-regime indicator that is small and statistically indistinguishable from zero (implied open-regime elasticity 1.43, $p = 0.55$ for equality). The fleet's production response to ecological conditions does not differ across regulatory regimes. The reduced-form elasticity is an equilibrium object combining the production technology (the $\gamma < 1$ catch function in the structural model) with the fleet's endogenous response through entry and exit; the regime-invariance licenses the structural separation of biology process from regulation that makes the climate counterfactuals tractable.

These facts map directly onto the structural objects the framework distinguishes. The choice response (extensive margin dominant) calibrates the nested-logit choice model and the choice diffuseness parameter σ in Section 5. The spatial displacement pattern (100–200 km, aggregate-neutral) is the empirical analogue of the leakage matrix in Section 2.3 and disciplines σ against the reduced-form reach. The bunching evidence validates $\xi_k > 0$ at access patches. The biomass elasticity establishes the responsiveness of harvest to ecological conditions and therefore the importance of uncertainty.

5 Structural model

We implement the framework empirically with a nested-logit choice model of fleet behavior, a size-structured biology, a calibrated catch production function, and a marginal-harvest perturbation instrument that traces the Control FOC. Three foundational data fixes give the exercise credibility: the SAMS-anchored biomass panel, a Berry-style calibration of baseline area shares, and an explicit defense of the choice-diffuseness parameter against its naive estimate. Detailed parameter values, validation against the NOAA assessment, and sensitivity exercises are in Appendix **??**.

Fleet behavior. A nested-logit choice model generates vessel-level location choice, with trips choosing over (area, trip-day) alternatives nested by spatial cluster. Behavioral parameters include the DAS shadow α, the revenue response β_R, the distance cost β_d, month-year fixed effects, and the nest-correlation parameter λ_{nest}. We use the estimated model in expectation, with aggregated choice probabilities at the boat-occasion level rather than per-boat Monte Carlo draws. The model maps directly to the vessel FOC in Section 2.3: the cap shadow $\xi_{k,t}$ and DAS shadow μ_t are the multipliers on observed catch caps and DAS allocations and rationalize the boats that are constrained by the caps in different access areas and the overall DAS cap.

Biology and production. Each patch carries a vector biomass over four size classes: pre-recruit, small (S2+), medium (S1), and u10 (S0, the highest-priced class). Von Bertalanffy growth with patch-specific parameters from the published NOAA assessment governs transitions across classes. Recruitment is log-normal at the patch level with patch-specific marginal variances and a cross-patch correlation structure Σ_ε recovered from biomass-survey residuals; the climate state θ_t enters through the mean, variance, and correlation of these shocks (Section 6.3). Mortality combines a natural rate $M = 0.1$ per year with fishing mortality from realized harvest. The catch production function is $\text{catch}_{k,t} = q \cdot B_{k,t}^{\gamma}$ with $\gamma = 0.40$ estimated from causal/depletion variation. The current version uses a common elasticity across size classes. Prices are observed and are used to estimate expected revenue from each patch k in week t, which are used in the choice model.

The SAMS-anchored biomass panel. The structural exercise requires a patch-level panel of in-water biomass that the fleet's marginal-harvest decisions can be priced against. We construct this from the published PDT Scallop Area Management Simulator (SAMS) combined-survey biomass estimates by management subarea, mapping each model patch to its nearest SAMS subarea and allocating the published total across patches by where the fleet actually fishes. These, combined with estimates from scallop dredge surveys, can separate out The panel delivers patch-level biomasses roughly consistent with the published PDT/SAMS totals and fleet exploitation rates ($F \approx 12\%$ of biomass caught). The biomass shocks are the residual from the expected growth parameters; these residuals are calculated from existing scallop dredge surveys and assembled in a covariance matrix of biomass shocks across space.

The marginal-harvest perturbation. The empirical counterpart of the Control FOC is computed by perturbation. In FY2014, we force every boat-trip at patch k to land an additional $\delta_{kg} = 100$ kg per trip-day; the fleet then re-optimizes its spatial allocation in subsequent years under the perturbed biology. We compute MC_k, the marginal cost of conservation in patch k or $\xi_{k,t}$, as the additional profit from this change in harvest. We compute MB_k from simulating forward how the perturbation changes future stocks and therefore future profits, inclusive of the substitution across space implied by the choice model. It is the negative of the discounted sum of future profit changes, with discount factor $\beta = 0.95$ and the horizon extending to FY2019. Monte Carlo draws over biology shocks ($n_{\text{draws}} = 100$ in the headline analysis).

The cost parameters. The choice model estimates area cost parameters (mean cost of traveling to any patch k) alongside the fixed cost of fishing in an area. We also estimate σ, the variance of the logit cost shocks, and estimate $\sigma = \$200$K—the headline value—the fleet's spatial substitution roughly matches the reduced-form displacement reach to 100–200 km documented in Section 4. The leakage term is highly sensitive to σ.

6 Results

We report the empirical exercise in three parts. We first document that the FY2014 regulation fails all three efficiency conditions at the margin and decompose the spatial misallocation across patches and boats. We then show how each condition responds to a different climate channel under projected mid-century biology. We close with the welfare-weighted analysis and the decoupling of static and dynamic conclusions under inequality aversion.

6.1 Testing the regulation in 2014

Static intratemporal efficiency: relating MB and MC overall. Summing MB and MC across all areas in 2014, the marginal benefit of conservation exceeds the marginal cost by about 63% (Table 3). A kilogram of scallop harvest at the margin carries an expected future stock-value loss of roughly \$34, against a dock value of \$21, i.e. a wedge of \$13/kg. At observed regulations, a scallop in the sea is still worth considerably more than landed. Overall, therefore, harvest limits and day caps set in 2014 are too lax if the goal was to maximize the profits of the scallop fishery in net present value. Table 6 shows that leakage destroys about 17% of the marginal benefit of conservation in any area: reducing harvests in an area on the margin causes some increased harvest elsewhere. Many of those areas have more plentiful scallops or are otehrrwise less profitable than the areas targeted for clsoure or limited harvests, and therefore the leakage is far from 1-to-1. Nevertheless, an alternative mix of spatial policies could improve the value of holding scallops in the sea.

Spatial efficiency: ratio of MB to MC over space. The aggregate masks substantial heterogeneity. Per-kg shadow values range from ∼\$8/kg at under-fished grounds (area_37, biomass 13M kg) to ∼\$485/kg at the most over-fished one (area_24, biomass 3.2M kg)—a span of ∼60× within the access regime, leading to wide ranges in the ratio of MB to MC across patches of the ocean (Table 4). Certain access areas with plentiful scallops, alongside some open areas with smaller biomass, are too restricted ($MC_k > MB_k$), while many smaller access areas could see even tighter limits or be completely closed. Meanwhile, all closed areas have scallops that are much more worthwhile on the margin in the sea than being harvested in 2014, given the possibility of future growth. In fact, the closures are areas with some of the highest marginal benefits of conservation and in that sense are well targeted. Areas near access areas, i.e. those most likely to see considerable substitution of harvest, are some of the most over-fished by this metric. This is due mostly to leakage from areas in the rotational program that leaves the stock highly degraded right outside.

Misallocation due to spatial caps or due to boats? The dispersion in patch-level wedges combines two conceptually distinct misallocations: the harvests in the patch (either directly regulated or implicitly using day restrictions) are not at the optimal level to preserve the stock, or the non-tradable DAS allocation and harvest limits in access areas means boats with different shadow prices fish the same grounds. We separate them by a variance decomposition of $\log \mathrm{MB}/\mathrm{MC}$ across boat-patch cells. Three quarters of the dispersion lives between patches; only about 10% lives between boats fishing the same patch (Table 5). The dominant source of the spatial inefficiency is therefore coarseness of the cap structure, not heterogeneity across vessels. While the regulation does overly restrict the most profitable boats— estimated boat fixed effects correlate with the vessel-level shadow price at $+0.97$—the decomposition suggests that policy tools often supported by economists such as landing taxes or tradeable harvest permits would not improve efficiency as much relative to changing overall harvest goals across areas.

Dynamic intertemporal efficiency: the asset-pricing residual. The third efficiency condition concerns whether the shadow value of stock evolves along the costate equation (5)—whether the regulation's implicit pricing of in-water stock tracks its forward asset-pricing value as the system moves through time. Along the dynamic optimum, the realized residual

$$u_{k,t+1} \equiv \beta\big(- C_{k,X}(q^*_{k,t+1}, X_{k,t+1}) + \lambda_{k,t+1}\big)\big(1 + G_{k,X_k}\big) - \lambda_{k,t}$$

has conditional expectation zero. We compute $\mathbb{E}[u]$ along the simulated 2014–2019 path under the status-quo regulation and the historical biology calibration; this involves solving for MC and MB in 2014 and again in future years. In the base case (the first row of Table 8), we find the average residual to be about \$4.6 million. A positive residual here suggests that the regulations implicitly under-value the growth of scallops as closures and access areas update year to year. Future work can understand the spatial dispersion in this value. The caveat, however, is that this is under existing resource growth parameters even as it changed.

6.2 Distributional incidence and the decoupling of static and dynamic

The empirical exercise above weights all vessels equally, as if the regulator's goal is to maximize total expected profit. The conservation gains from any of the reforms above do not, however, accrue equally across the fleet. New Bedford accounts for 41% of fleet-wide MC but 52% of fleet-wide MB—an 11-pp gain in share; the Mid-Atlantic accounts for 26% of MC but only 14% of MB, a 12-pp loss. The Mid-Atlantic is also characterized by smaller, less profitable vessels and lower-income communities. As a simple test of equity preferences affecting the results, we compute equity-weighted versions of both the static wedge and the dynamic residual under a CRRA-like aggregator $U(\pi) = \pi^{1-p}/(1-p)$ with weights $w_i \propto \text{revenue}_i^{-p}$. Table 8 reports both objects across the plausible range of p.

The static wedge crosses zero at about $p = 1.5$: a planner who puts more weight on dollars to lower-profit vessels will easily rationalize the existing stringency of harvests in 2014. The areas that are deemed over-fished are also those areas that are more often targeted by less profitable boats in the Mid-Atlantic. In addition, it compresses profits by not forcing Mid-Atlatnic boats to travel to farther, costlier locations in bad states when their less lucrative neighboring fishing grounds do not do well. The dynamic residual also shrinks under inequality aversion: the regulations still do not update in line with maximizing value but the consequences to an inequality-averse regulator are smaller. Aggregate tightening of the static wedge, the most salient reform under equal weighting, weakens or reverses under moderate inequality aversion.

6.3 Changing resource growth: the climate channels

We project the structural model to decompose the impact of different changes in the ecological conditions: observed changes in the mean growth of the resource, in the overall variance of biomass shocks, and in the spatial correlation of ecological shocks. We first simulated forward using the growth and covariance matrix of biomass from earlier years as if fixed innto the future (a stationary state). Then we compare that to the observed results from earlier. We can therefore see whether some of the observed inefficiency is due to under-reaction to a changing ocean.

The aggregate static wedge widens sevenfold, primarily through variance and covariance. MB/MC moves from 1.35 in the stationary system to 1.63. We attribute this almost entirely the entire widening to the variance and covariance (about equally), with the falling mean-growth contributing only little. The increasing uncertainty in growth and the increased spatial correlation of ecological shocks (scallops unexpectedly growing or dying together) increase the precautionary value of keeping the resource in the ocean.

The spatial dispersion widens due to changing ecology. The cross-patch variance of $\log \mathrm{MB}_k/\mathrm{MC}_k$ rises by $\sim$50% under the observed changing ocean, but the attribution is different. Variance alone widens the dispersion ($+0.32$); covariance alone *narrows* it (-0.17), because more synchronized shocks make patches' precautionary responses more uniform; their interaction is modest ($+0.11$); and the three-way interaction of mean, variance, and covariance is the largest single contribution to the widening ($+0.27$). Nevertheless,

The dynamic residual emerges from mean-growth degradation. Under a stationary climate, the realized residual $\mathbb{E}[u]$ is essentially zero ($-\$0.12$M, statistically indistinguishable from zero)—the status-quo regulation tracks the dynamic optimum reasonably well under historical conditions. Most of that widening is from the mean main effect; variance and covariance channels contribute under $0.3M each. The mechanism is the costate equation's persistence factor: degraded mean recruitment lowers G_{k,X_k}, raising the forward asset-pricing value of stock on the right-hand side of the costate equation, while the regulation's implicit shadow price (its left-hand side) holds at its historically calibrated level. This is rather striking evidence that the regulator does update but on old information on ecological growth.

The three failure modes do not require the same reform. A regulation that responds to climate by tightening the aggregate cap would close the intratemporal wedge but leave the dynamic and spatial dimensions untouched. A regulation that re-prices the stock—more frequent assessment, indexing of caps to projected biology, sunset clauses— would close the dynamic residual but not necessarily the others. Differentiated patch caps that respond to projected local variance and covariance would close the spatial dispersion. The three are not substitutes.

7 Discussion and Conclusion

The empirical exercise documents three distinct efficiency tests of the FY2014 spatial regulation. Each is structurally separable, responds to a different climate channel, and calls for a different reform.

Three implications extend beyond the Atlantic sea scallop fishery. First, each of the three conditions tests something different about the regulation, each is sensitive to a different moment of the underlying environmental process, and each fails independently of the others. Distinguishing them substantially narrows the reform space. The diagnostic structure can be applied to any setting in which a regulator faces spatial heterogeneity under environmental uncertainty—forest management with stochastic growth, pollution permits under uncertain damages, water rights under hydrological variability.

Second, the variance-decomposition methodology for reform priority delivers a quantitative ranking of reform levers when multiple instruments could in principle close a given inefficiency. In this fishery, it places differentiated patch caps approximately seven times above permit market-style reform by welfare attribution. The general lesson is that whether "differentiate caps across cells" or "allow trade across agents" is the higher-priority reform depends on whether the source of misallocation is between-cell or between-agent. The empirical decomposition tells you which; the prior assumption that one or the other dominates is replaceable by data.

Third, the decoupling of static and dynamic conclusions under inequality preferences is, to our knowledge, novel in spatial-resource settings. The static wedge crosses zero around moderate inequality aversion because the conservation gains accrue disproportionately to DAS-constrained high-revenue vessels; the dynamic residual stays positive in sign across the full plausible range of distributional preferences because the mean-growth channel that drives it propagates through the biology process rather than through the cross-vessel distribution of profit. We conjecture this decoupling appears in other applications where the static wedge is dominated by precautionary terms aggregating over agent-level profits and the dynamic residual is driven by mean-process channels propagating through the costate. If so, the practical implication generalizes: under distributional uncertainty, the reform direction most robust to planner choice differs from the reform direction most salient when just maximizing the NPV of a resource in each period.

There are some caveats for the future. In the current model, general-equilibrium price effects are assumed away, despite common downward-sloping demand being the canonical reason for leakage across space; the empirical exercise understates leakage by ignoring the price channel. A separate identification of demand variation would be needed to quantify this margin. In addition, the distributional analysis focuses only on the distribution of profits across boats in different ports, not at labor or community-level welfare concerns that would require more information on how resource rent flows to different groups.

The broader lesson is methodological. Spatial regulation under uncertainty is a domain in which the textbook treatment offers a strong conceptual narrative—a Hotelling-rule analog for renewable resources, augmented by the cross-cell structure of optimal allocation—that can be used to test for the value of spatial management under rapidly changing ecological conditions. The three types of efficiency tests offer distinct solutions and can later be used to suggest different reforms. Diagnosing spatial regulation under uncertainty benefits from the finer structural decomposition we have developed here. The identification of which efficiency condition is failing, which climate channel is amplifying it, and which reform addresses it is the work of the empirical exercise. The model offers tests that can be taken to a wide range of natural resources in a changign climate.

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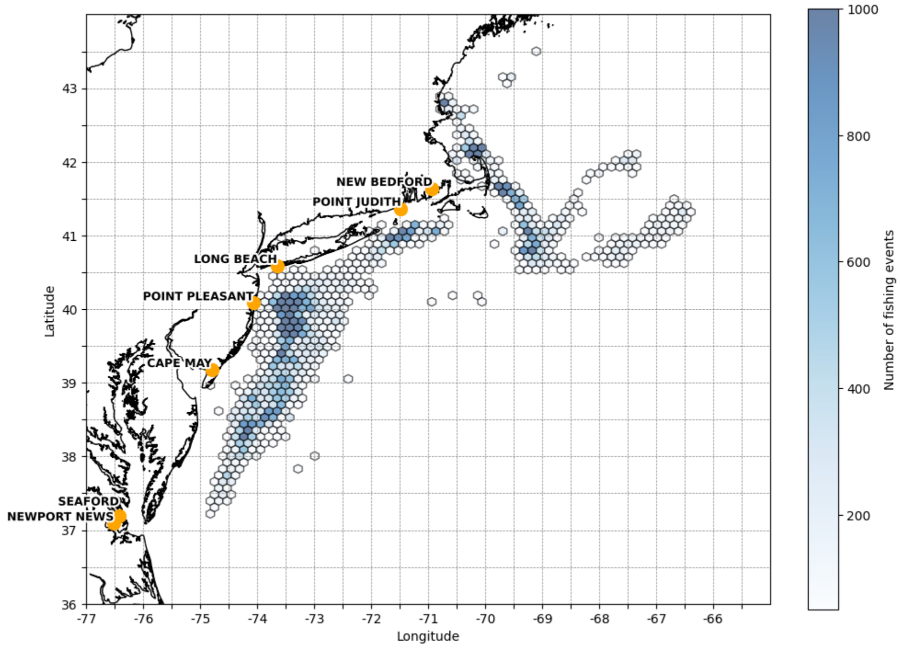
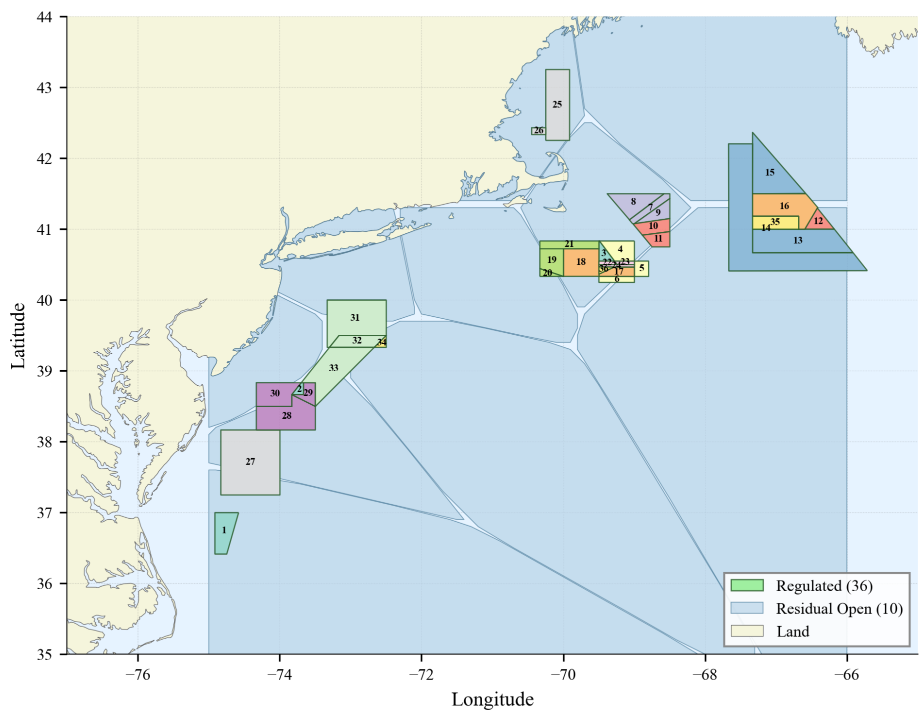
Figure 1. Map of areas



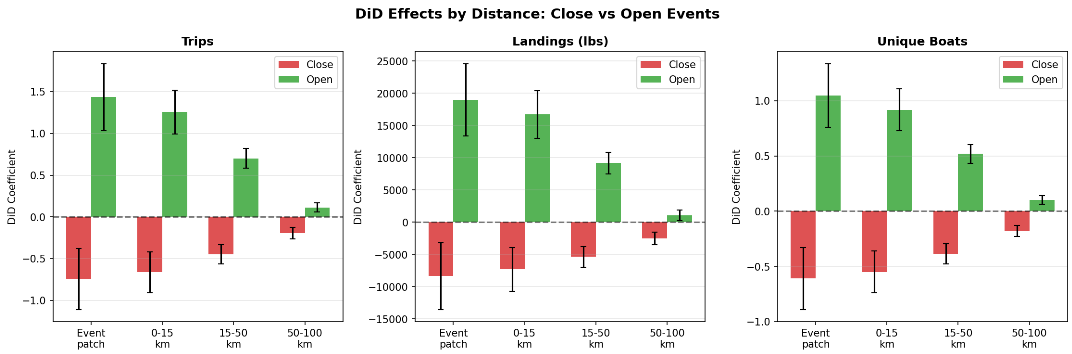
(a) Map of all harvests, 2008-2022



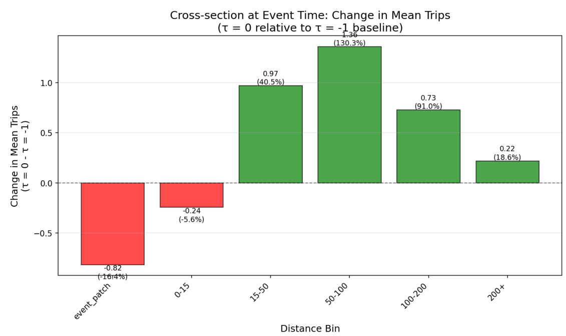
(b) Areas created for location-choice regression

Note: The figure shows two maps of the Northeast US scallop fishing grounds. Sub-figure (a) shows the distribution of harvests across all years, which are concentrated off of the coasts of New Jersey and Cape Cod. Sub-figure (b) summarizes our construction of areas used in the location-choice model.

Figure 2. Patch-level regression results: substitution across time



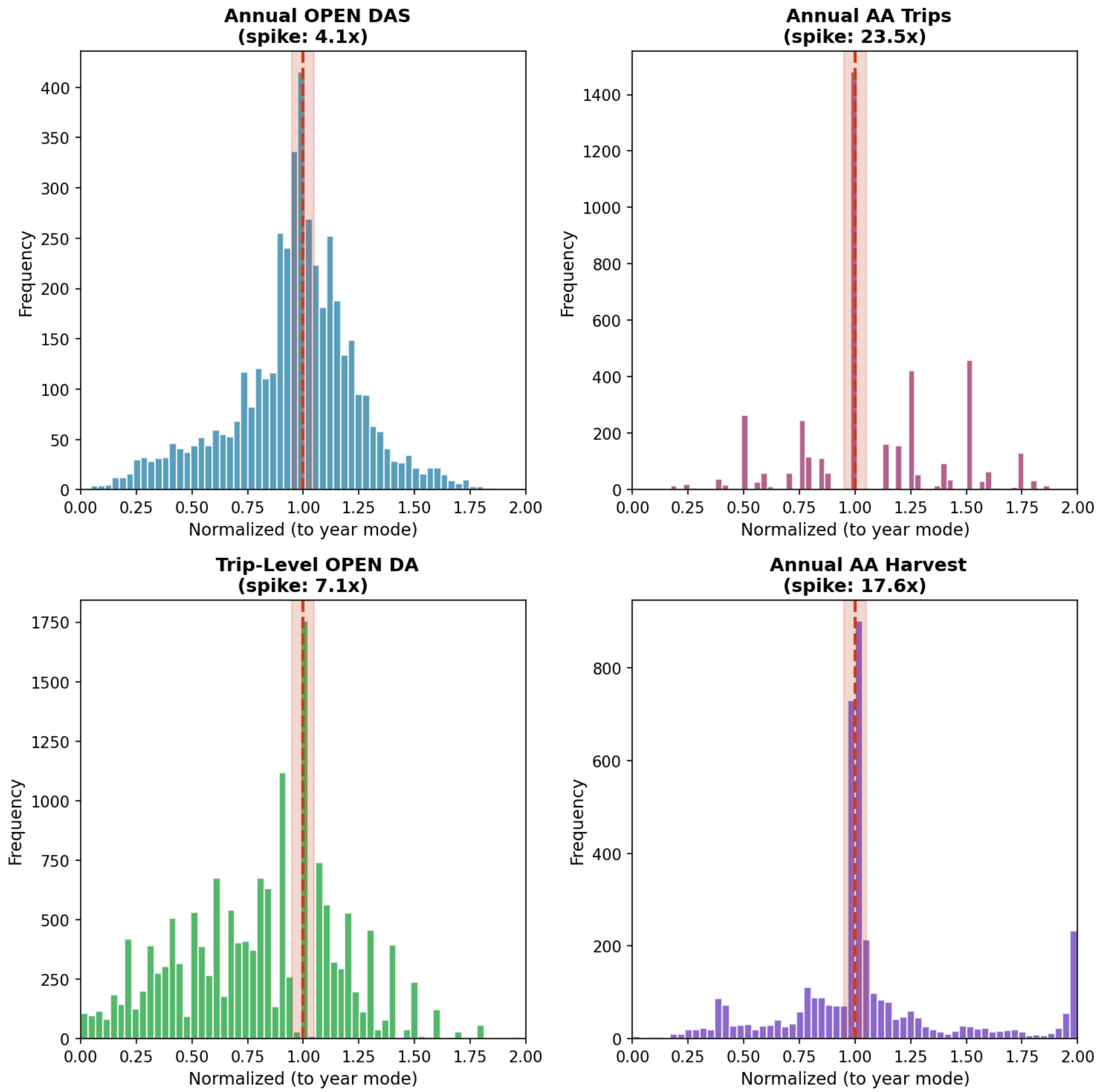
(a) Change in trips after closure by distance



(b) Change relative to areas 200+ km away

Note: This figure shows patch-level changes in fishing activity following a closure event, estimated using a difference-in-differences design. Panel (a) reports treatment effects by distance bin from the closed patch for three outcomes: number of trips, total landings (lbs), and number of unique vessels. Red bars correspond to the closed patch itself; green bars correspond to open patches at increasing distances (0–15 km, 15–50 km, 50–100 km). Coefficients are relative to pre-closure means in the same patches and year fixed effects; error bars denote 95% confidence intervals. Panel (b) presents the same trip effects at the event date, normalized relative to patches more than 200 km away. The figure shows that closures lead to large declines in activity in the treated patch and substantial displacement to nearby patches, with effects decaying rapidly in distance. This pattern indicates that spatial substitution is highly local.

Figure 3. Bunching comparison



Note: This figure compares bunching behavior around regulatory limits under different management regimes and outcome measures. The panels plot histograms of annual or trip-level outcomes normalized by the vessel-year mode (vertical dashed line at 1), so that values near one indicate observations at the most common regulatory limit. The reported "spike" indicates the ratio of the mass at the mode to the average mass in neighboring bins, capturing the strength of bunching at the constraint. The figure shows strong bunching in access-area trips and harvest, indicating that quantity limits bind tightly in practice, while effort limits in open areas generate substantially weaker bunching. This contrast implies that the binding constraints are the harvest controls in access areas, while days at sea measures are affected by measurement error and bunching is not as clear.

Table 1. Summary Statistics

	N	Mean	p5	p50	p95
Trip duration (hrs.)	68,001	73.4	8	22	276
# fishing events/trip	70,058	2.175	1	2	5
Quantity landed/trip (lbs.)	70,058	6,047	100	599	26,488
Value landed/trip ($)	70,058	57,819	1,066	6,879	254,638
Vessel # trips/year	4,635	31.2	2	16	128
Vessel crew	4,635	5.8	2	7	9
Vessel size (GT)	4,635	110	14	124	196
# vessels/year	9	515	490	500	597
# home ports/year	9	92.5	89	91	109
Frac. revenue to New Bedford		0.405			
Frac. revenue to Cape May		0.178			

Note: The table shows summary statistics from the detailed trip data available from the National Marine Fisheries Service. It aggregates across all years from 2008 to 2024. New Bedford and Cape May are the only two ports with more than 4% of the share of total revenue in the data. The first column is the sample size; the others are the mean and the 5th, 50th, and 95th percentiles respectively. Quantity landed aggregates pounds (lbs.) across all size bins in the data. Only vessels with a limited-access permit and using a scallop dredge are included. Gross tonnage (GT) is a measure of ship volume.

Table 2. Vessel-level spatial choice and margin response.

	(1) cond. logit	(2) ext. margin	(3) int. margin
distance (100 km)	-0.55^{***}	-0.41^{***}	-0.04
	(0.03)	(0.05)	(0.06)
closed	-0.68^{***}	-1.15^{***}	-0.09
	(0.04)	(0.07)	(0.08)
access	$+0.71^{***}$	$+0.60^{***}$	$+0.02$
	(0.04)	(0.06)	(0.07)
% effect closure vs. open	-49%	-69%	-9%
% effect access vs. open	$+103\%$	$+82\%$	$+2\%$
year FE	yes	yes	yes
N	4,102,873	4,102,873	287,194

Notes: Column (1): trip-level conditional logit; column (2): vessel-area-year extensive-margin logit; column (3): vessel-area-year Poisson conditional on participation. SE clustered at vessel-year. $^{***}p < 0.001$.

Table 3. Aggregate static-efficiency wedge, FY2014 access focal set.

	\$M	per kg
MC (marginal current value)	22.0	\$21/kg
MB (discounted future cost)	35.9	\$34/kg
wedge $= \mathrm{MC} - \mathrm{MB}$	-13.9	$-\$13$/kg
MB/MC	**1.63**	

Table 4. Cross-regulatory picture: MB/MC by area type.

Area type (FY2014 status)	MB/MC	reading
Access, large biomass ($B > 5$M kg)	0.4–3.7	mixed
Access, small biomass ($B < 5$M kg)	1.5–23.1	more over-fished
Border areas to the rotational program	3–206	most over-fished
Open (mid-Atlantic, mixed biomass)	0.65–62	highly mixed
Closed	1.6–106	keep closed

Notes: Ranges of MB/MC across the patches in each category. $\mathrm{MB}/\mathrm{MC} > 1$ indicates over-fishing in 2014.

Table 5. Boat $\times$ patch variance decomposition of static-efficiency objects.

object	between-boat	between-patch	residual	R^2	corr(α_{boat}, DAS shadow)
$\log \mathrm{MC}$	9.3%	75.2%	15.6%	84%	**+0.97**
$\log \mathrm{MB}$	19.7%	69.2%	11.1%	89%	+0.84
$\log(\mathrm{MB}/\mathrm{MC})$	10.7%	**73.3%**	16.0%	84%	+0.76

Notes: Two-way ANOVA on boat-patch cells with $\mathrm{MC} > 0$ and $\mathrm{MB} > 0$. Rightmost column is the Pearson correlation between the estimated boat fixed effect and each vessel's revealed shadow price.

Table 6. Decomposition of aggregate MB, FY2014 focal set.

component	\$M	share of MB
own-stock	42.0	—
leakage (spatial substitution)	-6.1	-17%
MB = own + leakage	35.9	

Table 7. Climate amplification (mean growth, changed variance overall, changed covariance) across the three efficiency conditions.

Efficiency condition	Climate impact	Dominant climate channel	Channel attribution
Static intratemporal (MB/MC)	$1.35 \to 1.63$	variance & covariance	variance main effect $+0.14$; covariance $+0.13$; mean $+0.01$
Static spatial (var $\log \mathrm{MB}_k/\mathrm{MC}_k$)	$1.48 \to 2.28$	mean growth & variance	3-way interaction $+0.27$; covariance alone NARROWS dispersion (-0.17)
Dynamic ($\mathbb{E}[u]$)	$-\$0.12\text{M} \to +\$4.55\text{M}$	mean growth	mean main effect +\$3.95M (87% of widening)

Notes: Decomposition of how efficiency of the regulation as viewed in FY2014. The comparison estimates the marginal cost and benefit of conservation across patches when the biomass growth and shoack covariance matrix from the early period (2010-2016) is fixed into the later one, versus allowing the growth parameters to change as observed. 2014 starting condition.

Table 8. Equity-weighted efficiency conditions under inequality aversion

	Static wedge		Dynamic residual	Leaky bucket (90/10)
Parameter p	MB/MC	wedge \$M	$\mathbb{E}[u]$ \$M	\$ worth burning per \$1 delivered
0 (equal)	1.63	-13.9	$+4.55$	\$0.00
0.5	1.38	-9.4	$+3.81$	\$1.08
1 (log)	1.16	-3.6	$+3.21$	\$3.34
1.5	1.01	-0.2	$+2.66$	\$8.04
2	0.92	$+1.9$	$+2.21$	\$17.82
3	0.78	$+5.4$	$+1.62$	\$80.64
4	0.66	$+8.9$	$+1.27$	\$353.18

Notes: Static wedge under historical biology; dynamic residual under degraded climate. Welfare weights $w_i \propto \pi_i^{-p}$. The rightmost column reports the Okun-style leaky-bucket value implied by each p to move a dollar from the top to the bottom of the income distribution ($r = \pi_{90}/\pi_{10} = 4.34$), the dollar amount the planner would destroy to transfer \$1 from the 90th-percentile vessel to the 10th-percentile vessel. Generally, a regulator with parameter p should be willing to lose $1 - (\pi_L/\pi_H)^p$ to redistribute $\$1$ from percentile H to percentile L, where π_Q is the profit at percentile Q.

A Framework derivations

This appendix contains the derivations deferred from Section 2. We work through the vessel problem, the leakage matrix derivation via the implicit function theorem, the regulator's recursive problem and its two first-order conditions, the algebraic decomposition of the static shadow value, the closed-form steady-state approximation of the costate, the size-structured generalization to a vector of size classes, the welfare-weighted extension, and the general-equilibrium price treatment. Notation throughout matches Section 2.

A.1 The vessel's problem in full

The representative fleet at time t takes the regulatory policy $I_t = (\{\bar{Q}_{k,t}\}_{k\in\mathcal{A}}, \bar{T}_t)$ and the state X_t as given and solves

$$\max_{\{q_k\}_{k=1}^{K}} \sum_k \pi_{k,t}(q_k, X_{k,t}) \quad \text{s.t.} \quad q_k \leq \bar{Q}_{k,t} \text{ for } k \in \mathcal{A}, \quad \sum_k T_k(q_k) \leq \bar{T}_t, \quad q_k \geq 0, \tag{6}$$

with $\pi_{k,t}(q_k, X_k) = p_t q_k - C_k(q_k, X_k)$, $C_{k,q} > 0$, $C_{k,qq} > 0$, $C_{k,X} < 0$, and $T_k(q_k)$ the trip-day function mapping harvest at k into days consumed. With multipliers $\xi_{k,t}, \mu_t \geq 0$ on the cap and aggregate-effort constraints, the Lagrangian is

$$\mathcal{L}_t = \sum_k \pi_{k,t}(q_k, X_{k,t}) - \sum_{k\in\mathcal{A}} \xi_{k,t}\,(q_k - \bar{Q}_{k,t}) - \mu_t \left(\sum_k T_k(q_k) - \bar{T}_t\right),$$

and the patch-level first-order condition is

$$p_t = C_{k,q}(q_k^*, X_{k,t}) + \xi_{k,t} \cdot \mathbb{1}[k \in \mathcal{A}] + \mu_t\, T_k'(q_k^*), \tag{7}$$

with complementary slackness $\xi_{k,t} \cdot (q_k^* - \bar{Q}_{k,t}) = 0$ and $\mu_t \cdot (\sum_k T_k(q_k^*) - \bar{T}_t) = 0$.

The cap rent $\xi_{k,t}$ is the marginal value to the fleet of cap relaxation: by the envelope theorem applied to the optimized profit at patch k,

$$\frac{\partial \pi_{k,t}^*}{\partial \bar{Q}_{k,t}} = \xi_{k,t}. \tag{8}$$

The DAS shadow μ_t is the analogous rent on the aggregate effort constraint: $\partial \sum_k \pi_{k,t}^*/\partial \bar{T}_t = \mu_t$. These two shadow values—one at each access area, one fleet-wide on aggregate effort—are the objects the empirical exercise uses as inputs.

A.2 The leakage matrix via the implicit function theorem

Treat the vessel FOC system (7) as a system of K equations in K unknowns $\{q_k^*\}$, indexed by the cap configuration $\{\bar{Q}_{k,t}\}_{k\in\mathcal{A}}$ and the aggregate effort allowance $\bar{T}_t$. We are interested in how the optimal harvest allocation responds to a change in the cap at one access area k_0, holding the rest of the regulatory policy fixed.

At the constrained patch $k = k_0 \in \mathcal{A}$, the cap binds and $q_{k_0}^* = \bar{Q}_{k_0,t}$ trivially, so

$$\frac{\partial q_{k_0}^*}{\partial \bar{Q}_{k_0,t}} = 1. \tag{9}$$

For all other patches $j \neq k_0$, the FOC at j does not directly involve $\bar{Q}_{k_0,t}$, but it involves μ_t which is itself a function of all caps. Differentiate (7) at j:

$$0 = (C_{j,qq}(q_j^*, X_j) + \mu_t T_j''(q_j^*)) \cdot \frac{\partial q_j^*}{\partial \bar{Q}_{k_0,t}} + T_j'(q_j^*) \cdot \frac{\partial \mu_t}{\partial \bar{Q}_{k_0,t}}. \tag{10}$$

The first term on the right captures the direct effect of q_j^* moving on the marginal cost and the trip-day function; the second captures the indirect effect through μ_t moving. Solving (10) for the cross-patch derivative:

$$\boxed{\frac{\partial q_j^*}{\partial \bar{Q}_{k_0,t}} = -\frac{T_j'(q_j^*)}{C_{j,qq}(q_j^*, X_j) + \mu_t T_j''(q_j^*)} \cdot \frac{\partial \mu_t}{\partial \bar{Q}_{k_0,t}}.} \tag{11}$$

Two observations follow. First, the cross-patch derivative is proportional to $\partial\mu_t/\partial\bar{Q}_{k_0,t}$: in the absence of the aggregate effort constraint ($\mu_t \equiv 0$ and $\partial\mu_t/\partial\bar{Q}_{k_0,t} = 0$), the cross-patch derivatives are identically zero and there is no leakage. The leakage matrix exists *because of* the binding aggregate effort constraint. Second, the sign: relaxing the cap at k_0 allows more harvest there, which consumes more days at sea, tightening the aggregate effort budget. The shadow value μ_t rises: $\partial\mu_t/\partial\bar{Q}_{k_0,t} > 0$. Given $T_j' > 0$ and the convex-cost denominator positive, the cross-patch derivative is negative for $j \neq k_0$. Effort at every other patch falls in response to cap relaxation at k_0. This is the canonical leakage direction; tightening the cap reverses the sign.

The full leakage matrix has the structure

$$\frac{\partial q_j^*}{\partial \bar{Q}_{k_0,t}} = \begin{cases} 1 & \text{if } j = k_0, \\ -T_j'(q_j^*)\big/\big(C_{j,qq} + \mu_t T_j''\big) \cdot \partial\mu_t/\partial\bar{Q}_{k_0,t} & \text{if } j \neq k_0. \end{cases}$$

Aggregating across j, the conservation of days-at-sea pins down $\partial\mu_t/\partial\bar{Q}_{k_0,t}$ implicitly through

$$T'_{k_0}(q^*_{k_0}) + \sum_{j \neq k_0} T'_j(q^*_j) \cdot \frac{\partial q^*_j}{\partial \bar{Q}_{k_0,t}} = 0$$

when $\bar{T}_t$ binds. Substituting (11) and solving for $\partial \mu_t / \partial \bar{Q}_{k_0,t}$ closes the system.

A.3 The regulator's recursive problem and its first-order conditions

The regulator chooses the sequence of policies $\{I_t\}$ to maximize expected discounted welfare. Let $W(X_t, I_t) = \sum_k \pi^*_{k,t}(X_t, I_t)$ be the within-period welfare flow under risk-neutral, equally-weighted aggregation (the welfare-weighted generalization is in Appendix A.7). The value function satisfies the Bellman equation

$$V(X_t) = \max_{I_t} \big\{ W(X_t, I_t) + \beta\, \mathbb{E}_t[V(X_{t+1})] \big\}, \tag{12}$$

with $X_{k,t+1} = G_k(X_{k,t}, q^*_{k,t}, \varepsilon_{k,t+1})$ and $\varepsilon_{t+1} \sim F_\varepsilon(\cdot; \theta_t)$.

Control FOC. Differentiate (12) with respect to $\bar{Q}_{k_0,t}$ for $k_0 \in \mathcal{A}$. The within-period derivative invokes the envelope condition (8): $\partial W / \partial \bar{Q}_{k_0,t} = \xi_{k_0,t}$. The continuation derivative is, by chain rule,

$$\beta\, \mathbb{E}_t \left[\sum_j V_{X_j}(X_{t+1}) \cdot \frac{\partial X_{j,t+1}}{\partial \bar{Q}_{k_0,t}} \right].$$

Define the costate $\lambda_{j,t+1} \equiv V_{X_j}(X_{t+1})$, and use $\partial X_{j,t+1} / \partial \bar{Q}_{k_0,t} = G_{j,Q} \cdot \partial q^*_j / \partial \bar{Q}_{k_0,t}$ from the stock transition. Setting the total derivative to zero:

$$\boxed{\xi_{k_0,t} = -\beta\, \mathbb{E}_t \left[\sum_j \lambda_{j,t+1} \cdot G_{j,Q} \cdot \frac{\partial q^*_j}{\partial \bar{Q}_{k_0,t}} \right].} \tag{13}$$

The left-hand side is the cap rent (the within-period marginal cost of cap relaxation, equivalently the marginal benefit of harvest today the fleet would forgo if the cap were tightened); the right-hand side is the discounted future stock-value benefit of conservation, summed across patches through the leakage matrix (11) and the depletion derivative $G_{j,Q}$.

Costate equation. Differentiate the Bellman with respect to $X_{k,t}$. The regulator has optimized I_t, so by the envelope theorem the indirect effect through I_t vanishes. Using

$\partial W/\partial X_{k,t} = \partial \pi^*_{k,t}/\partial X_{k,t} = -C_{k,X} > 0$ (the marginal cost saving from more stock at k):

$$\boxed{\lambda_{k,t} = -C_{k,X}(q^*_k, X_{k,t}) + \beta\,\mathbb{E}_t\left[\sum_j \lambda_{j,t+1} \cdot \left(G_{j,X_k} + G_{j,Q} \cdot \frac{\partial q^*_j}{\partial X_{k,t}}\right)\right].} \tag{14}$$

The first term is the static piece: an extra unit of stock at k saves $-C_{k,X}$ in current cost. The second is the recursive piece, with two channels: biology persistence (G_{j,X_k}, how stock at k today maps into stock at j tomorrow through the biological process) and the fleet's response to current stock through the within-period optimization ($G_{j,Q} \cdot \partial q^*_j/\partial X_{k,t}$).

Equations (13) and (14) are structurally distinct. The Control FOC is an optimality condition on the regulator's chosen cap, holding for each access area separately. The Costate equation is a law of motion for the shadow value of stock, describing how $\lambda_{k,t}$ should evolve over time conditional on the regulator's having intratemporally optimized. The standard Euler equation is the substitution of (14) into the future term of (13), which collapses the two into a single expression at the cost of hiding the distinction. The empirical exercise tests the two conditions separately.

A.4 Algebraic decomposition of the static shadow value

The Control FOC (13) involves the expectation of a product. Apply the identity $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y] + \text{Cov}(X, Y)$ to the term inside the bracket:

$$\mathbb{E}_t\left[\sum_j \lambda_{j,t+1} \cdot G_{j,Q} \cdot \frac{\partial q^*_j}{\partial \bar{Q}_{k_0,t}}\right] = \sum_j \mathbb{E}_t[\lambda_{j,t+1}] \cdot G_{j,Q} \cdot \frac{\partial q^*_j}{\partial \bar{Q}_{k_0,t}} + \sum_j \text{Cov}_t\left(\lambda_{j,t+1},\, G_{j,Q} \cdot \frac{\partial q^*_j}{\partial \bar{Q}_{k_0,t}}\right).$$

(We treat $G_{j,Q}$ as evaluated at the conditional expectation and $\partial q^*_j/\partial \bar{Q}_{k_0,t}$ as a deterministic function of (X_t, I_t), so the covariance lives entirely in $\lambda_{j,t+1}$.) Substituting into (13) and rearranging signs using $\partial q^*_{k_0}/\partial \bar{Q}_{k_0,t} = 1$, $\partial q^*_j/\partial \bar{Q}_{k_0,t} < 0$ for $j \neq k_0$, and $G_{j,Q} < 0$ throughout:

$$\xi_{k_0,t} = \underbrace{\beta\,\mathbb{E}_t[\lambda_{k_0,t+1}] \cdot |G_{k_0,Q}|}_{\text{(a) own-patch saving}} - \underbrace{\beta \sum_{j\neq k_0} \mathbb{E}_t[\lambda_{j,t+1}] \cdot |G_{j,Q}| \cdot \left|\frac{\partial q^*_j}{\partial \bar{Q}_{k_0,t}}\right|}_{\text{(b) leakage}} + \underbrace{\beta \sum_j \text{Cov}_t\left(\lambda_{j,t+1},\, -G_{j,Q} \cdot \frac{\partial q^*_j}{\partial \bar{Q}_{k_0,t}}\right)}_{\text{(c) precautionary covariance}}. \tag{15}$$

The signs of the three terms follow from the sign analysis just noted. Term (a) is positive: own-patch future stock-value benefit foregone is the discounted expected future shadow value times the depletion rate. Term (b) carries a negative sign in the equation but represents a benefit (across-patch effort reallocation preserves stock at $j \neq k_0$), so the sign convention reflects that this benefit *offsets* the conservation cost being charged at k_0. Term (c) is the precautionary covariance and captures the second-moment contribution to the wedge: under deterministic biology, term (c) is identically zero. The empirical -17% leakage share is the ratio (b)/[(a)−(b)+(c)] in absolute value.

A.5 Closed-form steady-state shadow value

Under stationary policy and stationary biology, in a steady state where $\lambda_{k,t} = \lambda_{k,t+1} \equiv \lambda_k$ and with cross-patch dependencies in the costate set to zero (no biological dispersal in the persistence channel, no cross-patch dependence of harvest on stock through the choice model), the Costate equation (14) collapses to

$$\lambda_k = -C_{k,X}(q_k^*, X_k) + \beta \cdot \lambda_k \cdot G_{k,X_k}(X_k, q_k^*).$$

Solving for λ_k:

$$\boxed{\lambda_k \;\approx\; \frac{-C_{k,X}(q_k^*, X_k)}{1 - \beta\, G_{k,X_k}(X_k, q_k^*)}.} \tag{16}$$

The numerator is the static piece: current marginal cost-saving from extra stock. The denominator $(1-\beta G_{k,X_k})$ is the inverse capitalization factor—a Faustmann-style discount that weights the infinite future stream of cost savings by the persistence of stock. The approximation requires $\beta G_{k,X_k} < 1$; if violated, the shadow value diverges in steady state and the approximation breaks down. Every object on the right-hand side is estimable from the vessel choice model ($C_{k,X}$ evaluated at observed q^* and X) and the biological model (G_{k,X_k}), without solving a dynamic-programming problem or imposing a terminal condition.

For climate comparative statics, the closed-form expression shows the channels through which θ_t acts on λ_k. A mean shift in θ_t moves G_{k,X_k} through expected biology, acting on the denominator. A variance or covariance shift in θ_t does not enter the closed-form expression directly (which is deterministic) but acts through the stochastic terms in the full Costate equation (14)—this is why the dynamic-efficiency residual is dominated by the mean-growth channel empirically.

A.6 Size-structured generalization

The framework in the main text treats X_k as a scalar. For Atlantic sea scallops the relevant stock object is a vector of size classes. Replace the scalar X_k with a vector

$$\mathbf{X}_k = (X_k^{\text{pre}}, X_k^{\text{small}}, X_k^{\text{med}}, X_k^{\text{u10}})^\top,$$

where the four entries are pre-recruit, small (S2+), medium (S1), and u10 (S0, the highest-priced class) biomasses respectively. The stock transition becomes

$$\mathbf{X}_{k,t+1} = \mathbf{G}_k(\mathbf{X}_{k,t}, q_{k,t}, \varepsilon_{k,t+1}),$$

with $\mathbf{G}_k$ a vector-valued transition function. The within-period profit function becomes $\pi_{k,t}(q_k, \mathbf{X}_k) = \mathbf{p}_t^\top \mathbf{q}_k - C_k(q_k, \mathbf{X}_k)$ where $\mathbf{p}_t$ is the size-class-specific price vector (with u10 commanding the highest price) and $\mathbf{q}_k$ is the size-class composition of the catch.

The Costate equation generalizes to a vector recursion. Let $\boldsymbol{\lambda}_{k,t} = \nabla_{\mathbf{X}_k} V(X_t)$ and let $\mathbf{G}_{k,X}^{\text{class}}$ denote the Jacobian of the size-structured transition with respect to $\mathbf{X}_k$: $(\mathbf{G}_{k,X}^{\text{class}})_{m,m'} = \partial X_{k,t+1}^{(m)} / \partial X_{k,t}^{(m')}$, with positive off-diagonal entries capturing maturation (small $\to$ medium $\to$ u10). Then

$$\boldsymbol{\lambda}_{k,t} = -\mathbf{C}_{k,X} + \beta\, \mathbb{E}_t\big[(\mathbf{G}_{k,X}^{\text{class}})^\top \boldsymbol{\lambda}_{k,t+1}\big] \tag{17}$$

(suppressing cross-patch dependencies as in the closed-form treatment above). Under stationarity the closed-form generalization is a matrix inversion:

$$\boldsymbol{\lambda}_k \;\approx\; (\mathbf{I} - \beta\, \mathbf{G}_{k,X}^{\text{class}})^{-1}\, (-\mathbf{C}_{k,X}). \tag{18}$$

Two structural facts follow immediately. First, $\boldsymbol{\lambda}_k$ inherits a maturation premium: the shadow value of a marginal small scallop is augmented by the discounted probability that it matures into the medium and u10 classes, weighted by the prices of those classes. Working through the matrix inversion, the shadow value of class m at patch k is the discounted sum, over time horizons h, of the cost-savings from that scallop being in each class at time $t+h$ weighted by the maturation probabilities along the path. Second, the own-patch term (a) in the decomposition (15) carries this maturation premium structure; the leakage term (b) is unchanged in structure because maturation operates within a patch, not across patches.

The empirical signature of this maturation channel is the hump-shaped impulse response of profit to a marginal harvest: the per-period profit loss peaks one to two years after the perturbation rather than in the year of perturbation, because the harvested cohort would have entered higher-priced size classes over that horizon. This pattern is qualitatively impossible in a scalar-stock model and is direct empirical evidence for the size-structured costate.

A.7 Welfare-weighted extension

Replace the equal-weighted welfare aggregator $W = \sum_k \pi_k^*$ with a weighted CRRA aggregator

$$W = \sum_i w_i \, U(\pi_i^*), \qquad U(\pi) = \pi^{1-p}/(1-p), \qquad U'(\pi) = \pi^{-p}, \tag{19}$$

where i indexes vessels (extending the vessel-heterogeneous setup in Section A.8 below) and the weights w_i are chosen by the planner. For the planner with inequality aversion $p \geq 0$, natural weights are $w_i \propto \text{revenue}_i^{-p}$, placing larger weight on lower-revenue vessels.

The Control FOC becomes

$$\xi_{k_0,t} \cdot \overline{U'} = -\beta \, \mathbb{E}_t \left[\sum_j \lambda_{j,t+1} \cdot G_{j,Q} \cdot \frac{\partial q_j^*}{\partial \bar{Q}_{k_0,t}} \right] \tag{20}$$

where $\overline{U'} = \sum_i w_i U'(\pi_i^*) \cdot$ (share of vessel i's revenue at patch k_0) is the welfare-weighted average marginal utility at the perturbed patch. The right-hand side picks up the analogous welfare weights through the future $\lambda_{j,t+1}$ via the aggregation across vessels of future profits. The structural form of the decomposition (15) is preserved: terms (a), (b), and (c) still have their respective economic content; what changes is the dispersion of vessel-level weights across patches.

Two implications follow. First, the static wedge is planner-specific. A redistributive planner (higher p) weights the DAS-constrained high-revenue vessels less heavily. If the conservation gains from cap tightening accrue disproportionately to those vessels—as the empirics in Section 6 document—the welfare-weighted $\overline{U'}$ in the right-hand side shrinks more than the left-hand side, and the welfare-weighted wedge crosses zero at some $p > 0$.

Second, the Costate equation reweights symmetrically through $-C_{k,X}$ and the forward $\lambda_{j,t+1}$ on both sides of (14). The dynamic residual along the regulated path is therefore reweighted by the welfare weights but does not pick up the asymmetry that the Control FOC does. The static and dynamic efficiency conditions decouple along the equity dimension: the static case is planner-specific while the dynamic-efficiency failure is planner-invariant in sign. The empirical exercise documents this decoupling at the level of magnitudes: the static wedge crosses zero around $p \approx 2$ while the dynamic residual remains positive at every plausible inequality-aversion parameter.

A.8 Vessel heterogeneity

The framework as developed above uses a representative fleet. Extending to a heterogeneous fleet indexed by $i = 1, \dots, N$ preserves the structural form of the decomposition. Vessel-specific cost functions $C_{i,k}(q_{i,k}, X_k)$, vessel-specific trip-day functions $T_{i,k}$, and vessel-specific DAS constraints with shadow values $\mu_{i,t}$ replace the representative versions. The patch-level cap $\bar{Q}_{k,t}$ is split across vessels through the institutional allocation rule; let $\bar{Q}_{i,k,t}$ denote vessel i's share.

The vessel-level FOC at (i, k) becomes

$$p_t = C_{i,k,q}(q^*_{i,k}, X_k) + \xi_{i,k,t} \cdot \mathbb{1}[k \in \mathcal{A}] + \mu_{i,t}\, T'_{i,k}(q^*_{i,k}), \tag{21}$$

and the cross-patch leakage matrix at the vessel level is

$$\frac{\partial q^*_{i,j}}{\partial \bar{Q}_{i,k_0,t}} = -\frac{T'_{i,j}(q^*_{i,j})}{C_{i,j,qq} + \mu_{i,t} T''_{i,j}} \cdot \frac{\partial \mu_{i,t}}{\partial \bar{Q}_{i,k_0,t}},$$

identical in structure to (11) but at the vessel level. Aggregate $\xi_{k,t}$ and μ_t are weighted aggregates of vessel-level shadow values, with weights given by each vessel's share of harvest or effort at the relevant patch or fleet-wide.

The empirical implementation in Section 6.1 identifies the vessel-level structure through a two-way fixed-effect decomposition. The boat fixed effect on the log of the per-cell shadow value isolates the vessel-specific $\mu_{i,t}$ component—the cross-vessel dispersion in DAS shadow—and is correlated at +0.97 with the directly observed DAS shadow at the vessel level. The patch fixed effect isolates the patch-level structure of $\xi_{k,t}$, reflecting the coarseness of the patch-level cap structure.

A.9 General-equilibrium prices

The framework as developed treats the ex-vessel price p_t as exogenous. A natural extension is to make it endogenous through an inverse demand function $p_t = P(\sum_k Q_{k,t})$, where $Q_{k,t}$ is aggregate harvest at patch k and P is the inverse demand for the resource at the dock.

The Control FOC picks up a price-channel leakage term in addition to the DAS-channel leakage analyzed above. Differentiating the vessel FOC with p_t now a function of total harvest:

$$P\left(\sum_k Q_{k,t}\right) + Q_{k,t} \cdot P'\left(\sum_k Q_{k,t}\right) = C_{k,q} + \xi_{k,t} + \mu_t T'_k,$$

and the IFT analysis of Section A.2 now needs to account for p_t moving when $\bar{Q}_{k_0,t}$ moves. The result is a cross-patch derivative

$$\left.\frac{\partial q_j^*}{\partial \bar{Q}_{k_0,t}}\right|_{GE} = \left.\frac{\partial q_j^*}{\partial \bar{Q}_{k_0,t}}\right|_{DAS} + \frac{P'(\cdot)}{C_{j,qq} + \mu_t T''_j} \cdot \frac{\partial \sum_k Q_{k,t}}{\partial \bar{Q}_{k_0,t}},$$

with the second term capturing the price-channel substitution: when total harvest moves, the price moves, which moves harvest at every patch.

The price channel reinforces the substitution direction implied by the DAS channel. Tightening the cap at k_0 reduces aggregate harvest, which raises p_t through inverse demand, which raises the marginal product of effort at every patch j, which raises harvest at j. The two channels (DAS and price) push in the same direction. The two can be separated empirically only with variation in demand independent of supply regulation; the empirical analysis in the paper holds p_t fixed and so subsumes the price-channel leakage into the structural leakage term, understating total leakage relative to a full GE treatment.